

江西路泰通电子有限公司

产品规格承认书 Specification

品名/Product Name: LCM 模组

机种名/Product Number: LTT024A006

客户名称/Customer Name: 海能达

客户项目/Project Name: P50

版本号/ Revision No. VER.01

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客户工程确认/Customer R&D Approved by: _____

客户品质确认/Customer Quality Approved by: _____

Prepared by	Checked by	Approved by

Revision Sheet 变更单

Remark/备注:

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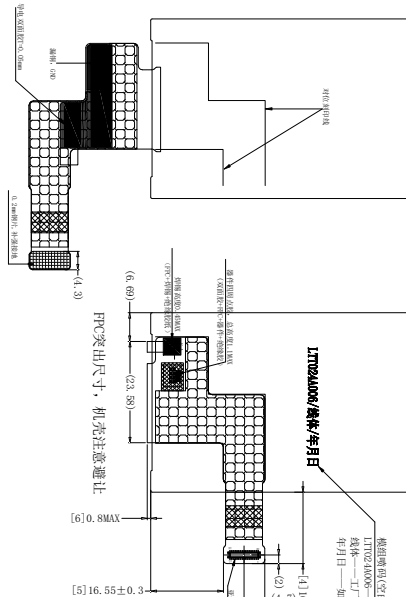
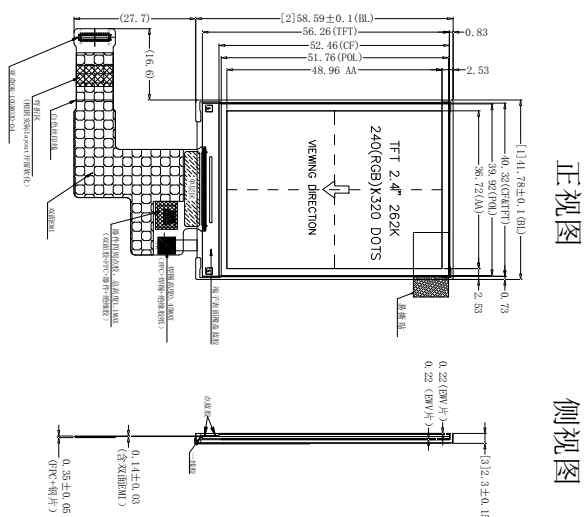
1. General Information

No.	Item	Specification	Unit
1	Panel Size	2.4"	inch
2	Number of Pixels	240 RGB(H)*320(V)	pixels
3	Active Area	36.72(H)*48.96(V)	mm
4	Pixel Pitch	0.153(H)*0.153(V)	mm
5	Outline Dimension	41.78(W)*58.59(H)*2.3(T)	mm
6	Number of Colors	262K	-
7	Display Mode	Normally White	-
8	Viewing Direction	12 O' CLOCK	-
9	Pixel Arrangement	RGB stripe	-
10	Luminance (cd/m^2)	500min,550 TYP	nit
11	Contrast Ratio	200min,500TYP	
12	Interface	SPI	-
13	Backlight	White LED	-
14	Diver IC	ST7789P3	
15	Operation Temperature	-20~70	°C
16	Storage Temperature	-30~80	°C

2. External Dimensions (2D 图纸)

TFT NOTES:

1. DISPLAY TYPE: BOE 2.4 IN, Normally White;
2. VIEWING DIRECTION: $\pm 12^\circ$ clock;
3. Driver IC: ST7789P3-G6
4. STORAGE TEMPERATURE: -30°C TO 80°C ,
OPERATING TEMPERATURE: -20°C TO 70°C ;
5. BL LED: 4 CHIP-WHITE LEDS, $I_f=20\text{mA}/\text{LED} \times 4$
LCM Luminance: $500\text{cd}/\text{m}^2$ (Min) ; $550\text{cd}/\text{m}^2$ (TYP)
White: $X=0.315 \pm 0.03$, $Y=0.325 \pm 0.03$
6. [] : CRITICAL DIMENSION; () : reference Dimensions.
7. GENERAL TOLERANCE: $\pm 0.1\text{mm}$;
8. RECOMMENDED CELLPHONE WINDOW SIZE: $0.2 \sim 0.3\text{mm}$ FROM LCD A.A AREA ;
9. 所有材料需符合RoHS2.0+REACH+H标准。



MARK	REV.	DESCRIPTION	REVISER	DATE
	AO	NEW	YL	2024.09.19
	AI	調整p10順序	YL	2025.03.01

★二级保密★

正视图

侧视图

背视图

FPC弯折示意图
弯折出货

LCM PIN

1	NC	30	GND
2	NC	29	NC
3	NC	28	NC
4	GND	27	GND
5	VCC	26	1.8V
6	VDDIO	25	1.8V
7	IO1	24	GND
8	IO2	23	NC
9	GND	22	NC
10	DIC	21	GND
11	CS	20	NC
12	SCL	19	NC
13	SDA	18	GND
14	SMD	17	GND
15	GND	16	75Ω

P50 TN 敏感 ID 接口:

ID 接口:

ID 模组出厂分配:

01:

软件ID: 0x87 0x87 0x99

```
graph TD
    ID1 --- R1[R1=10K]
    ID1 --- R3[R3=100K]
    ID1 --- VCC[VCC]
    ID2 --- R2[R2=10K]
    ID2 --- R4[R4=100K]
    ID2 --- GND[GND]
```

BACKLIGHT (CIRCUIT DIAGRAM)

VF=1.2~1.2V, I=2.0mA (恒定电流)

背光灯电路

4颗灯串联



江西路泰通电子有限公司

Inspect Field to tolerance:	± 0.20	UNITS: mm	DATE:		
DESIGN BY:			PART NUMBER : LT7024A006		SHEET : 1 OF 1
CHECKED BY:					
APPROVED BY:					DO NOT SCALE THIS DRAWING.
			DESCRIPTION:	MODEL	

3. Interface Description

Pin No.	Symbol	Description
1	NC	NC
2	NC	NC
3	NC	NC
4	GND	GROUND
5	VCC	Power Supply for Analog, Digital System and Booster Circuit.
6	VDDIO	Power Supply for I/O System.
7	ID1	Panel module identification 1.
8	ID2	Panel module identification 2.
9	GND	GROUND
10	D/C	Display data/command selection pin in serial interface.
11	CS	Chip selection pin.
12	SCL	Clock signal interface operation.
13	SDA	Serial in signal.
14	SDO	Serial out signal.
15	GND	GROUND
16	TE	Tearing effect output pin.
17	RESET	Reset pin.
18	GND	GROUND
19	NC	NC
20	NC	NC
21	GND	GROUND
22	NC	NC
23	NC	NC
24	GND	GROUND
25	LEDA	LED Anode.
26	LEDK	LED Cathode.
27	GND	GROUND
28	NC	NC
29	NC	NC
30	GND	GROUND

4. Absolute Maximum Ratings

Item	Symbol	Rating	Unit
Supply Voltage	VDD	- 0.3 ~ +4.6	V
Supply Voltage (Logic)	VDDI	- 0.3 ~ +4.0	V
Driver Supply Voltage	VGH-VGL	-0.3 ~ +30.0	V
Logic Input Voltage Range	VIN	- 0.3 ~ +4.0	V
Logic Output Voltage Range	VO	- 0.3 ~ +4.0	V
Operating Temperature Range	TOPR	-30 ~ +85	°C
Storage Temperature Range	TSTG	-40 ~ +125	°C

Table 1 Absolute Operation Range

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded.

Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

5. Electrical Characteristics

Parameter	Symbol	Condition	Specification			Unit	Related Pins
			MIN.	TYP.	MAX.		
Power & Operation Voltage							
System Voltage	VDD	Operating voltage	2.4	2.75	3.3	V	
Interface Operation Voltage	VDDI	I/O Supply Voltage	1.65	1.8	3.3	V	
Gate Driver High Voltage	VGH		12.2		14.97	V	Note 4
Gate Driver Low Voltage	VGL		-12.5		-7.16	V	
Gate Driver Supply Voltage		VGH-VGL	19.36		27.47	V	Note 5
Input / Output							
Logic-High Input Voltage	VIH		0.7VDDI		VDDI	V	Note 1
Logic-Low Input Voltage	VIL		VSS		0.3VDDI	V	Note 1
Logic-High Output Voltage	VOH	IOH = -1.0mA	0.8VDDI		VDDI	V	Note 1
Logic-Low Output Voltage	VOL	IOL = +1.0mA	VSS		0.2VDDI	V	Note 1
Logic-High Input Current	IIH	VIN = VDDI			1	uA	Note 1
Logic-Low Input Current	IIL	VIN = VSS	-1			uA	Note 1
Input Leakage Current	IIL	IOH = -1.0mA	-0.1		+0.1	uA	Note 1
VCOM Voltage							
VCOM amplitude	VCOM			VSS		V	
Source Driver							
Source Output Range	Vsout		VAN		VAP	V	
Gamma Reference Voltage(Positive)	VAP		4.45		6.4	V	Note 6
Gamma Reference Voltage(Negative)	VAN		-4.6		-2.65	V	
Source Output Settling Time	Tr	Below with 99% precision			20	us	Note 2
Output Offset Voltage	VOFFSET				35	mV	Note 3

Table 2 Basic DC Characteristics

Notes:

1. TA = -30 to 70°C (to +85°C no damage).
2. Source channel loading = 2KΩ + 12pF/channel, Gate channel loading = 5KΩ + 40pF/channel.
3. The Max. value is between measured point of source output and gamma setting value.
4. When evaluating the maximum and minimum of VGH, VDD = 2.8V.

6. Timing Characteristics.

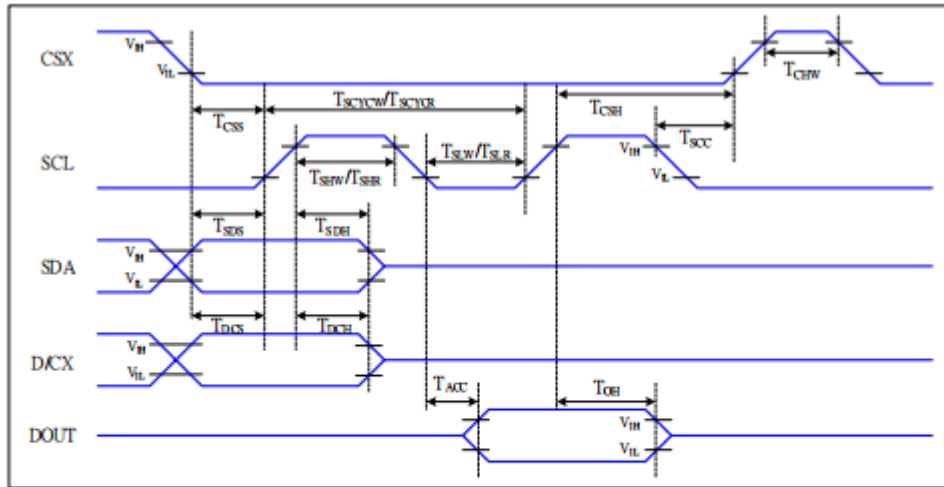


Figure 5 4-line serial Interface Timing Characteristics

VDD=1.65 to 3.3V, VDD=2.4 to 3.3V, AGND=DGND=0V, Ta=25°C

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T _{css}	Chip select setup time (write)	15	-	ns	
	T _{sch}	Chip select hold time (write)	15	-	ns	
	T _{css}	Chip select setup time (read)	60	-	ns	
	T _{scc}	Chip select hold time (read)	65	-	ns	
	T _{chw}	Chip select "H" pulse width	40	-	ns	
SCL	T _{scycw}	Serial clock cycle (Write)	16	-	ns	-write command & data ram
	T _{shw}	SCL "H" pulse width (Write)	7	-	ns	
	T _{slw}	SCL "L" pulse width (Write)	7	-	ns	
	T _{scycr}	Serial clock cycle (Read)	150	-	ns	-read command & data ram
	T _{shr}	SCL "H" pulse width (Read)	60	-	ns	
	T _{slr}	SCL "L" pulse width (Read)	60	-	ns	
D/CX	T _{dcs}	D/CX setup time	10	-	ns	
	T _{dch}	D/CX hold time	10	-	ns	
SDA (DIN)	T _{ds}	Data setup time	7	-	ns	
	T _{dh}	Data hold time	7	-	ns	
DOUT	T _{acc}	Access time	10	50	ns	For maximum CL=30pF
	T _{oh}	Output disable time	15	50	ns	For minimum CL=8pF

Table 6 4-line serial Interface Characteristics

Note1 : The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for input signals.

Note2 : In the read sequence of Serial interface, the 500nsec delay time is needed between read command and first read clock.

6.1 Reset Timing:

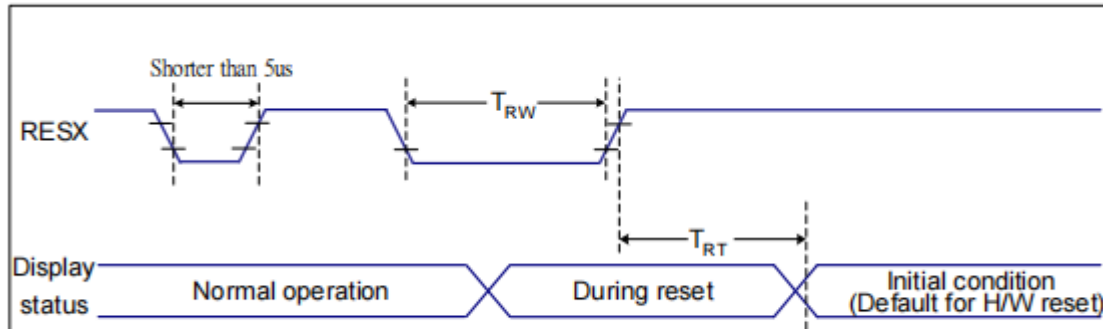


Figure 7 Reset Timing

$VDDI=1.65$ to $3.3V$, $VDD=2.4$ to $3.3V$, $AGND=DGND=0V$, $Ta=25^{\circ}C$

Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	TRW	Reset pulse duration	10	-	us
	TRT	Reset cancel	-	5 (Note 1, 5)	ms
			-	120 (Note 1, 6, 7)	ms

Table 8 Reset Timing

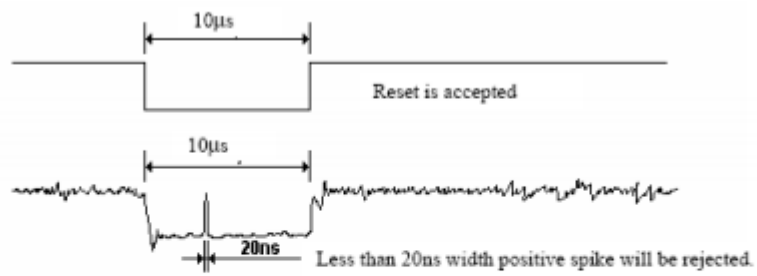
Notes:

1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (t_{RT}) within 5 ms after a rising edge of RESX.
2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out -mode. The display remains the blank state in Sleep In -mode.) and then return to Default condition for Hardware Reset.

4. Spike Rejection also applies during a valid reset pulse as shown below:



5. When Reset applied during Sleep In Mode.

6. When Reset applied during Sleep Out Mode.

7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

6.2 Power ON/OFF Sequence

VDDI and VDD can be applied in any order.

In CABC function application, VDDI power on need delay 5ms after VDD has been supplied.

VDD and VDDI can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VDD and VDDI must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDD can be powered down minimum 0msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

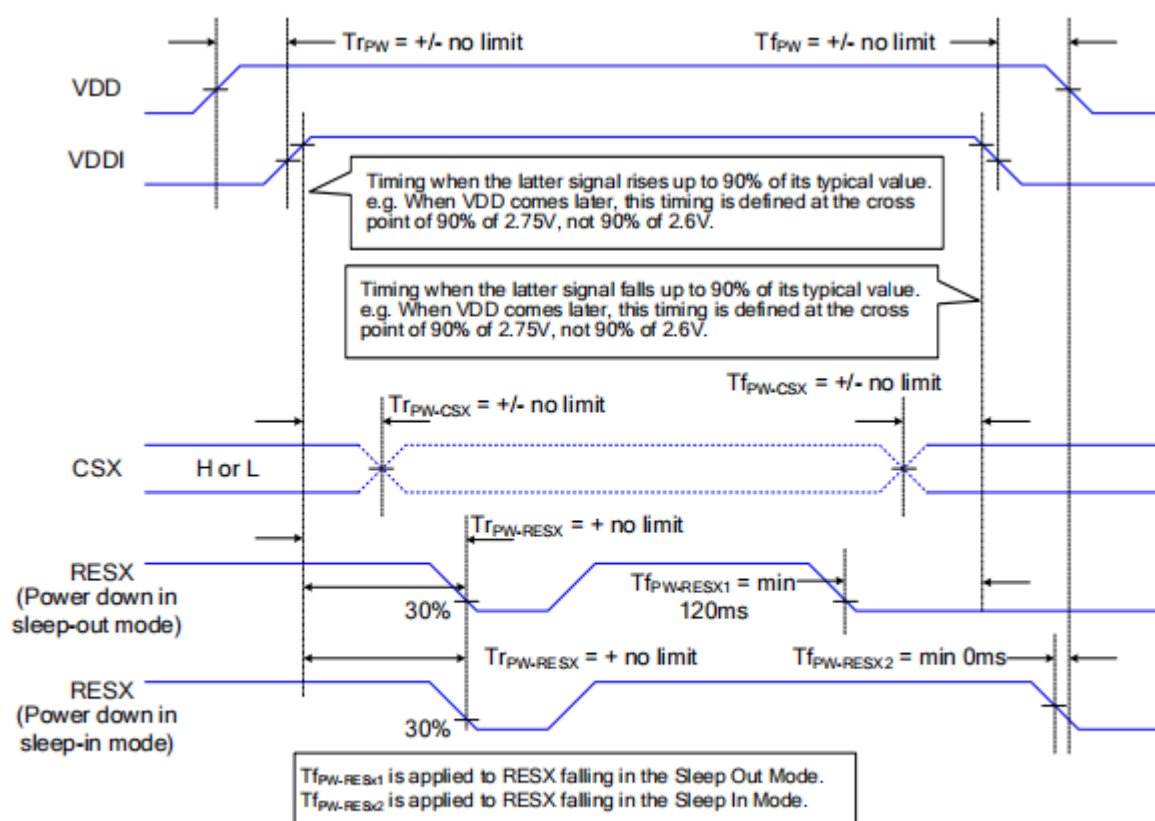
Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

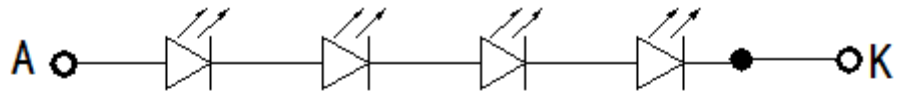
Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below



7. Backlight Characteristics.



BACKLIGHT (CIRCUIT DIAGRAM)

$$V_F = 11.2 \sim 12V \quad I_m = 20mA$$

背光电路 4颗灯串联

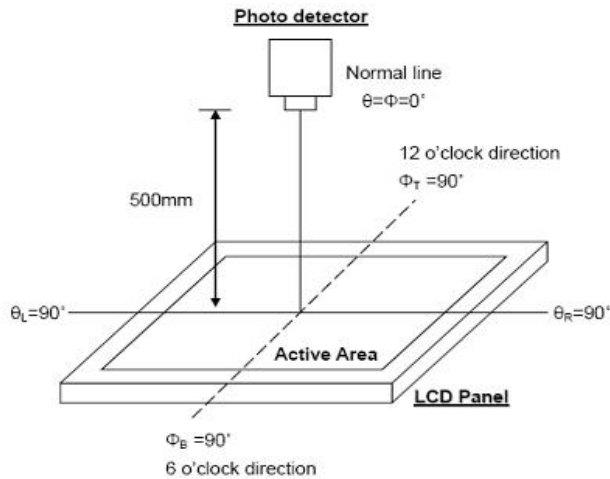
Item	Symbol	Min.	Typ.	Max.	Unit	Remark
LED current	I _{LED}	-	20	-	mA	4LEDS
Forward voltage	V _F	11.2	-	12	V	I _F =20mA 4LEDS
Reverse current	I _R	-	-	1	μA	V _R =5V, 1LED
Power dissipation	P _d	240			mW	4EDS
Peak forward current	I _{FP}	80			mA	1LED
Reverse Voltage	V _R	5			V	1LED

8. Optical Characteristics

Ta=25°C±2

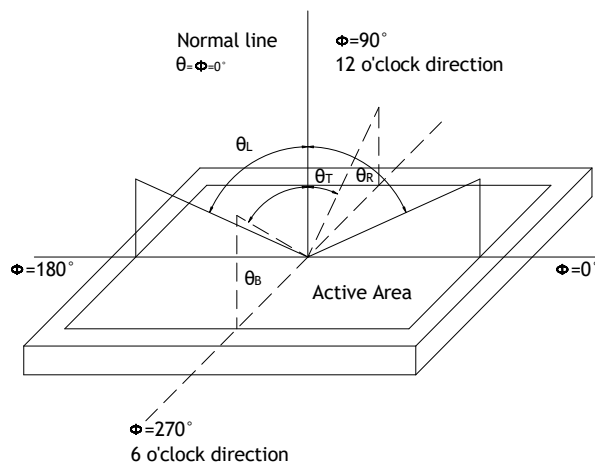
Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Contrast Ratio		CR	Θ = 0°	200	500	-		Note1 Note2
Luminance		Brightn ess		500	550	-	cd/m2	Note1 Note6 Note7
Flicker						-25	db	
gamma				1.9	2.2	2.5		
Crosstalk		CT		-	-	3	%	
Luminance Uniformity		IV-M		80	-	-	%	
Response Time	Rising +Falling	Tr +Tf	Ta= 25℃ Θ = 0°	-	25	35	ms	Note1 Note3
Viewing Angle range	Horizontal	ΘL	CR >10	35	45	-		Note2
		ΘR		35	45	-		
	Vertical	ΘU		35	40	-		
		ΘD		15	20	-		
Color Chromaticity	White	x	Θ=0 Normal Viewing angle	-0.03	0.31	+0.03		Note1 Note5 Note7
		y			0.325			
	Red	x		-0.03	0.6140	+0.03		
		y			0.3610			
	Green	x			0.3580			
		y			0.6040			
	Blue	x			0.1540			
		y			0.1080			
NTSC				50	55	-	%	

Note1: Definition of optical measurement system



Note2: Definition of viewing angle range and measurement system

Viewing angle is measured at the center point of the LCD by CONOSCOPE (ergo-80).



Note3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (TON) is the time between photo detector output intensity changed from 90% to 10%. And fall time (TOFF) is the time between photo detector output intensity changed from 10% to 90%.

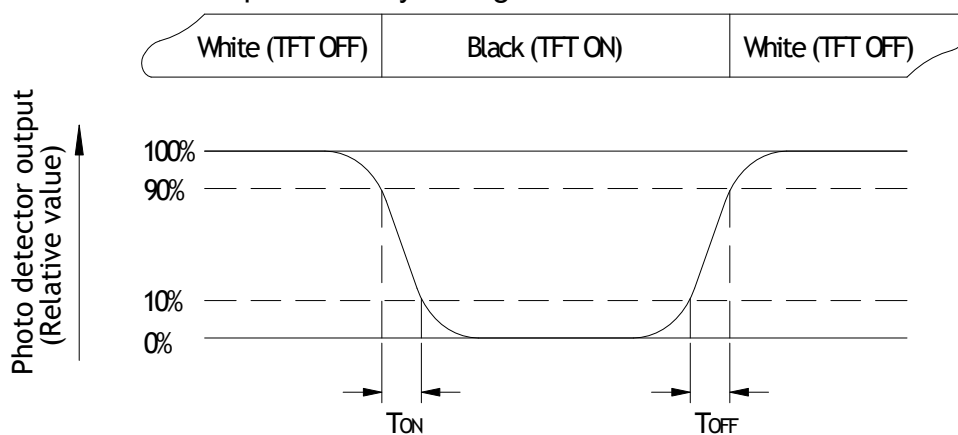


Fig. 6-3 Definition of response time

Note4: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD on the Whitestate}}{\text{Luminance measured when LCD on the Blackstate}}$$

“White state “: The state is that the LCD should drive by V_{white}.

“Black state”: The state is that the LCD should drive by V_{black}.

V_{white}: To be determined V_{black}: To be determined.

Note5: Definition of color chromaticity (CIE1931)

Color coordinates measured at center point of LCD.

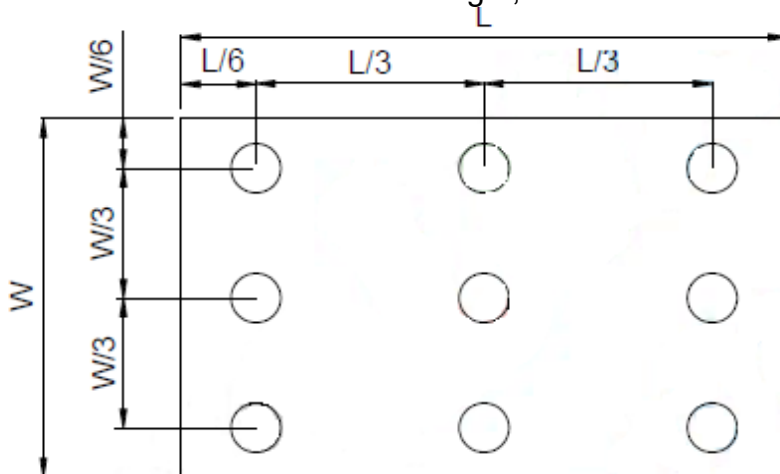
Note6: All input terminals LCD panel must be ground while measuring the center area of the panel. The LED driving condition is I_L=60mA

Note7: Definition of Luminance Uniformity

Active area is divided into 9 measuring areas. Every measuring point is placed at the center of each measuring area.

$$\text{Luminance Uniformity (U)} = L_{\min} / L_{\max}$$

L----Active area length, W---- Active area width



B_{max}: The measured maximum luminance of all measurement position.

B_{min}: The measured minimum luminance of all measurement position.

9. Reliability Test Conditions

NO.	TEST ITEMS	TEST CONDITION	INSPECTION AFTER TEST
①	High Temperature Storage	Ta =80°C,240H	Inspection after 4hours storage at room temperature, the samples should be free from defects: 1,Air bubble in the LCD. 2,Sealleak. 3,Non-display. 4,Missing segments. 5,Glass crack. 6,Current IDD is twice higher than initial value. 7, The surface shall be free from damage. 8,The electric Characteristics requirements shall be satisfied.
②	Low Temperature Storage	Ta= - 30°C;240H	
③	High Temperature Operation	Ta=70°C; 240 H	
④	Low Temperature Operation	Ta=-20°C; 240 H	
⑤	High Temperature and Humidity	Ta=60°C, 90%RH, 240 H	
⑥	Temperature Cycle(Storage)	-30°C(0.5h) ~ 80°C(0.5h) / 100 cycles	
⑦	ESD Test	Condition:R=330Ω,C=150PF, Air Mode:±8KV,10times Contack Mode:±4KV,10times	

REMARK: 备注

1, The Test samples should be applied to only one test item.

测试样例应该只应用于一个测试项目

2, Sample side for each test item is 3~5pcs.

样品每个项目 3~5pcs.

3, In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judge as a good part.

静电损坏造成的故障，如果在自然状态下重置能恢复过来，判断为良品。

4, EL evaluation should be excepted from reliability test with humidity and temperature: Some defects such as black spot/blemish can happen by natural chemical reaction with humidity and Fluorescence EL has.

EL 评价应该从与湿度和温度可靠性测试除外：一些缺陷如黑点/缺陷可以通过自然发生化学反应与湿度和荧光 EL。

5, Failure Judgment Criterion: Basic Specification Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

失败的判断标准：基本规范电气特性、机械特性、光学特性。

10. Inspection Item Condition

※参照《LCM+CTP IIS 海能达》标准, mura 参照 ND5%。



LCM+CTP IIS 海
能达.pdf

11. Handling Precautions

11.1 Mounting method

The LCD panel of quality Co.,ltd module consists of two thin glass plates with polarizes which easily be damaged. And since the module in so constructed as to be fixed by utilizing fitting holes in the printed circuit board.

Extreme care should be needed when handling the LCD modules.

11.2 Caution of LCD handling and cleaning

When cleaning the display surface, Use soft cloth with solvent [recommended below] and wipe lightly

- Isopropyl alcohol
- Ethyl alcohol

Do not wipe the display surface with dry or hard materials that will damage the polarizer surface.

Do not use the following solvent:

- Water
- Aromatics

Do not wipe ITO pad area with the dry or hard materials that will damage the ITO patterns

Do not use the following solvent on the pad or prevent it from being contaminated:

- Soldering flux
- Chlorine (Cl) , Salfur (S)

If goods were sent without being sili8con coated on the pad, ITO patterns could be damaged due to the corrosion as time goes on.

If ITO corrosion happen by miss-handling or using some materials such as Chlorine (Cl), Salfur (S) from customer, Responsibility is on customer.

11.3 Caution against static charge

The LCD module use C-MOS LSI drivers, so we recommended that you:

Connect any unused input terminal to Vdd or Vss, do not input any signals before power is turned on, and ground your body, work/assembly areas, assembly equipment to protect against static electricity.

11.4 packing

- Module employ LCD elements and must be treated as such.
- Avoid intense shock and falls from a height.
- To prevent modules from degradation, do not operate or store them exposed direct to sunshine or high temperature/humidity

11.5 Caution for operation

- It is an indispensable condition to drive LCD's within the specified voltage limit since the higher voltage then the limit cause the shorter LCD life.
- An electrochemical reaction due to direct current causes LCD's undesirable deterioration, so that the use of direct current drive should be avoided.
- Response time will be extremely delayed at lower temperature then the operating temperature range and on the other hand at higher temperature LCD's how dark color in them. However those phenomena do not mean malfunction or out of order with LCD's, which will come back in the specified operation temperature.
- If the display area is pushed hard during operation, some font will be abnormally displayed but it resumes normal condition after turning off once.
- A slight dew depositing on terminals is a cause for electro-chemical reaction resulting in terminal open circuit.

Usage under the maximum operating temperature, 50%Rh or less is required.

11.6 storage

In the case of storing for a long period of time for instance, for years for the purpose or replacement use, the following ways are recommended.

- Storage in a polyethylene bag with the opening sealed so as not to enter fresh air outside in it . And with no desiccant.
- Placing in a dark place where neither exposure to direct sunlight nor light's keeping the storage temperature range.
- Storing with no touch on polarizer surface by the anything else.
[It is recommended to store them as they have been contained in the inner container at the time of delivery from us

11.7 Safety

- It is recommendable to crash damaged or unnecessary LCD's into pieces and wash off liquid crystal by either of solvents such as acetone and ethanol, which should be burned up later.
- When any liquid leaked out of a damaged glass cell comes in contact with your hands, please wash it off well with soap and water

12. Precaution For Use

12.1

A limit sample should be provided by the both parties on an occasion when the both parties agreed its necessity. Judgment by a limit sample shall take effect after the limit sample has been established and confirmed by the both parties.

12.2

On the following occasions, the handing of problem should be decided through discussion and agreement between responsible of the both parties.

- When a question is arisen in this specification
- When a new problem is arisen which is not specified in this specifications
- When an inspection specifications change or operating condition change in customer is reported to quality Co.,Ltd , and some problem is arisen in this specification due to the change
- When a new problem is arisen at the customer's operating set for sample evaluation in the customer site.



13 Packing Method

TBD